

Task 'Introduction to Cloud Fundamentals and Services' Unit 8

Disaster Recovery Plan for Microsoft Azure Protected Cloud

Architecture

The suggested disaster recovery plan architecture is hosted on Microsoft Azure and consists of the following components:

Component	Purpose / Function
Ubuntu Linux Virtual Machine (VM)	Hosts the web application and PostgreSQL database. Stores configuration files, uploaded documents, and application data.
PostgreSQL Database	Stores structured application data and is hosted on the Ubuntu VM.
Azure Virtual Network (VNet)	Provides private networking and connectivity between the application components.
Azure Load Balancer	Distributes incoming application traffic to the VM/application services.
Azure Blob Storage	Stores application files such as uploaded documents and potentially backup data.
Primary Azure Region	Hosts the production application infrastructure.
Secondary Azure Region	Provides an alternate Azure location for disaster recovery.
Hourly PostgreSQL Dumps	Database backups generated every hour.

The secondary Azure region provides a disaster-recovery capability and the selected tool Restic, is an open-source backup tool, which supports encryption, deduplication of snapshots and the restoration of individual files or complete directory structure.

(Restic, 2026a).

Restic can use Microsoft Azure Blob Storage as a backup repository and can authenticate using Azure AD RBAC or some additional methods.

(Restic, 2026b).

Backup Strategy

Task 'Introduction to Cloud Fundamentals and Services' Unit 8

Restic runs hourly using Ubuntu's systemd scheduled timer and application files and database dumps are backed up into a dedicated Blob Storage container. Backups are encrypted before stored in the storage, and credentials are being protected within Azure Key Vault. Daily, weekly and monthly retention policies provide recorded recovery points. snapshots are replicated to a second Restic repository on the secondary Azure region on a regular basis.

The Blob Storage backup account is using a more restrictive RBAC permissions and protected using separated administrative account on a non-production VM. This reduces the risk that ransomware compromising the application VM can also delete backups. Repository integrity is periodically verified using restic check, and quarterly recovery drills validate that backups are usable.

(Restic, 2026a).

Failure Scenario and Recovery

The simulated scenario is a ransomware disaster file loss. At 10:00, an attacker compromises the application VM and caused encryption and deletion of stored customer documents. The incident is detected and verified at 10:15, and shortly after that it is being declared. The affected VM is then isolated from the network. The staff has identified the last known-good Restic snapshot created shortly before the encryption by the attacker, instead of backing up the compromised files. The known-good files are recovered onto a clean recovery VM using Restic's restore function and the database integrity checking is checked against the corresponding PostgreSQL dump. The

Task 'Introduction to Cloud Fundamentals and Services' Unit 8

restored data is then being reconnected by application and production is being restored using a consistent set of data. Restic also supports restoring either complete snapshots or selected files, making it suitable for both granular and full recovery.

(Restic, 2026a).

Recovery Objectives and Evaluation

Recovery Point Objective (RPO) is the maximum acceptable amount of data loss calculated in time, whereas Recovery Time Objective (RTO) is the target period for restoring the service after a disaster occurred.

(Microsoft, 2025).

For this DRP, the target RPO is 1 hour, and the RTO is 2 hours. During a simulated drill, the most recent backup was done 38 minutes ago, producing an RPO of 38 minutes. The newly deployed VM, data restoration, database validation and application restart took 74 minutes. This resulted in both targets being achieved.

In case of an Azure full regional outage, Azure Backup supports VM restoration from cross-region, recovery times and secondary-region RPO need to be considered for the chosen backup design.

(Microsoft, 2026).

Recovery exercises should integrate these speculations into DRP plans to ensure that such assumptions are being validated.

References

Task 'Introduction to Cloud Fundamentals and Services' Unit 8

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Task 'Introduction to Cloud Fundamentals and Services' Unit 8

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